Chadi Mountassir

Data Scientist | AI Engineer

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Portfolio / Projects

Profile

Data Scientist specializing in the development of high-impact, data-driven solutions for engineering and industrial environments. Strong expertise in machine learning, statistical analysis, and artificial intelligence. Currently working on optimizing mechanical part generation by integrating Generative AI models with CAD software, enabling intelligent automation and engineering innovation.

Education

- **National School of Applied Sciences of Al Hoceima (ENSAH)** — Morocco
  Engineering Degree – Data Engineering — *2020 – 2025*
- **Sijilmassa High School** — Morocco
  Baccalaureate in Mathematics Sciences A – Honors — *2019 – 2020*

Professional Experience

- **Capgemini Engineering** — Morocco
  Data Scientist – Hyper Automation & AI — *February 2025 – Present*
  - **Generative AI & CAD Integration**: Design and optimization of Generative AI–based solutions for mechanical part generation, modification, and optimization, fully integrated with industrial CAD software.
  - **Computer Vision for Technical Drawing Analysis**: Development of a computer vision model for automated detection and comparison of 2D drawing views from engineering files, accelerating design review and improving accuracy in technical documentation processing.
  - **Unified AI Agent Platform for Design Tools**: Design and development of a unified, agent-powered platform providing teams with natural language access to internally developed design tools, enabling automated execution of design-related tasks through conversational requests.
  - **Data & KPI Automation**: Implementation of data pipelines and automated workflows for computing, analyzing, and reporting critical engineering and supply chain KPIs.
  - **Production Deployment & Kubernetes**: End-to-end deployment of data and AI applications on the company's dedicated Kubernetes infrastructure, ensuring scalability, reliability, and secure integration with existing enterprise systems.
- **CastorPay** — Morocco
  Data Scientist — *July 2024 – September 2024*
  - **Fraud Detection**: Built and evaluated fraud detection models using large-scale transaction data.
  - **Model Industrialization**: Deployed machine learning models via secure REST APIs using Spring Boot and Docker to ensure scalable production environments.

Selected Projects

- **Private Cloud Infrastructure (Jan. 2025)**: Designed and deployed a private cloud infrastructure on local server. *OpenStack, Ubuntu Server, VMware*
- **Text-to-SQL – Generative AI Model (May 2024)**: Developed a Transformer-based model translating natural language into executable SQL queries, enabling automated data access and analysis. *PyTorch, HuggingFace, Streamlit*
- **Patent Data Analysis & Dashboarding (May 2024)**: Extracted, transformed, and visualized large-scale international patent data to support strategic and data-driven decision-making. *Selenium, Spark, MongoDB, Snowflake*

Skills

- **Programming**: Python, Java, SQL, Bash
- **Machine Learning & AI**: Scikit-learn, PyTorch, TensorFlow, Keras, Transformers, Hugging Face, ONNX, MLflow, Computer Vision (OpenCV), NLP (spaCy, NLTK), Generative AI, RAG, Fine-tuning, Prompt Engineering
- **Agentic AI & LLM Orchestration**: LangChain, LangGraph, LangFlow, CrewAI, AutoGen, LlamaIndex, OpenAI API, Azure OpenAI, Semantic Kernel, Vector Databases (Pinecone, Weaviate, ChromaDB, FAISS)

- **Big Data & Data Engineering**: Apache Spark, PySpark, Kafka, Hadoop (HDFS, YARN, MapReduce), Hive, Airflow, Flink, Databricks, Delta Lake, dbt, Data Lakes, ETL/ELT Pipelines
- **Databases**: MySQL, SQL Server, Oracle, PostgreSQL, MongoDB, Neo4j, Cassandra, Redis, Elasticsearch
- **Cloud & DevOps**: AWS (S3, EC2, EMR, SageMaker, Lambda, Glue), Docker, Kubernetes, Git, CI/CD, Terraform, OpenStack, CloudStack
- **Automation**: UiPath, Selenium, Robot Framework, Intelligent Document Processing (IDP)

Languages

Arabic (Native) French (Fluent) English (Fluent)

Leadership & Activities

- HR Lead – DATA-AI Club ENSAH (2023 – 2024)
- President – ENSAH Debate Club (2022 – 2023)
- Active Member – ENSAH Debate Club (2021 – 2022)